Getting started with spreadsheets.

1. Basic Computation
   (a) To add 3 + 4, enter `=3+4`
   (b) To subtract 100-76, enter `=100-76`
   (c) To multiply 4 times 18, enter `=4*18`
   (d) To divide 0.05 by 12, enter `=0.05/12`
   (e) To calculate 5^{25}, enter `=5^25`
2. Compute an 18% tip on a $35.75 bill.
   (a) What is 18% as a decimal? _____
   (b) What should we enter into the spreadsheet? _____
3. Make a tip calculator using **cell references**.
   (a) Open a new spreadsheet and label it `TIP`.
   (b) In cells A1, B1, C1, D1, type column headers:

Bill Amount	Tip Percent	Tip Paid	Total

   (c) In cells A2, B2, C2, D2, type:

=35.75	=0.18	=A2 * B2	=A2 + C2

   (d) What calculation did we do?

   (e) Use the calculator to determine the Total if the Bill Amount is $76.80 and the Tip Percent is 20%.

4. Keeping the Tip Percent at 20%, we now demonstrate the **Fill Down** feature.
   (a) Enter 10 into cell `A2` and enter =`A2`+10 into cell `A3`
   (b) Select `A3` and drag down until you get to 110.
   (c) Select cells `B2` through `D2` and drag the blue dot in the right-hand corner down.
   (d) How much is the tip on a $110 meal? _____

   How much is the final bill? _____

   How large does the Bill Amount need to be for the tip to be more than $15? _____

5. Quick Summary of Finance Terms

   (a) **principal**

   (b) **interest**

   (c) **interest rate**

   (d) **simple interest**

   (e) **accumulated amount** or **end amount** or **principal plus interest** or **future value**

   (f) **Formula for Simple Interest over Time**

   (g) Comment on time t.

6. Make a Simple Interest Calculator:

   Suppose \$1000 is invested in a savings account that earns 3.2% interest on the \$1000 each year.

   (a) Open a new spreadsheet/tab called `Interest`. Fill in the spreadsheet cells:

	A	B	C	D	E
1	Interest Rate	Principal	Year	Interest	Total
2					
3					

   (b) Use the pull down feature to determine the Total in the savings account in 10 years. Write down your steps.

   (c) How long will it take for the savings account to double in value? Write down your steps.

   (d) How can you calculate the Total in the savings account in 50 years **without** using the pull down feature?

   If P is the principal, r is the interest rate, t is the number of years, and A is the accumulated amount of money in the savings account, write a formula for A. Then check your calculation using the pull down feature.

   (e) A better simple interest calculator, using fixed cell references (the \$ fixes the row/column)

	A	B	C	D	E
1	Interest Rate	Principal	Year	Interest	Total
2	0.032	1000	0		=B2
3			=C2+1	= A2*B2	=D3+E2

   Now select and pull down `C3, D3, E3`. What happens if you change the interest rate to 4%? What if you change the principal to \$3000?

7. How To Export the **entire workbook** as a **PDF**.

   (a) Save your work!

   (b) In upper left menu, click `File` to produce a dropdown menu.

   (c) In dropdown menu, select `Download`

   (d) In right-side menu, select `PDF`